ATHANASIOS GOURDOMICHALIS
Java Programms
App5.java

Write an application that accepts an integer as **a command-line argument** when executed and checks whether this number belongs to the Fibonacci sequence or not. By definition, the first two Fibonacci numbers are 0 and 1, and each subsequent number is the sum of the previous two:

$$F_n = F_{n-1} + F_{n-2}$$

with $F_0 = 0$ and $F_1 = 1$

The following example of running the application is given with the number **121** as input data – the data is passed as an argument to the main method upon execution:

```
C:\Users>java App5 121
Fibonacci number = 0
Fibonacci number = 1
Fibonacci number = 1
Fibonacci number = 2
Fibonacci number = 3
Fibonacci number = 5
Fibonacci number = 8
Fibonacci number = 13
Fibonacci number = 21
Fibonacci number = 34
Fibonacci number = 55
Fibonacci number = 89
Fibonacci number = 144
121 is not a fibonacci number
```

The following execution example is also given with the number 8 as input data:

```
C:\Users>java App5 8
```

```
Fibonacci number = 0
Fibonacci number = 1
Fibonacci number = 1
Fibonacci number = 2
Fibonacci number = 3
Fibonacci number = 5
Fibonacci number = 8
8 is a fibonacci number
```



Complete the Java code in the file **App5.java**

Every program must compile and run from the command line (not through an IDE).